

ISyE 6740 – Spring 2026

Project Proposal

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Project Title: Air Quality Sensor Anomaly Detection and Correction

Problem Statement

Society relies on accurate air quality data to fight against air pollution and protect wellbeing, but deploying dense and commercial-grade monitoring infrastructure is cost-prohibitive for most cities. To bridge the gap, cities usually deploy low-cost sensor arrays. However, these devices are inherently noisy and prone to systemic failures, including calibration drift, hardware malfunctions, and environmental cross-sensitivity. Without robust and automated anomaly detection, faulty sensor readings blend into the dataset. These localized errors can severely skew predictive models, trigger false public safety alerts, and drive biased policy decisions.

Research showed that detecting and correcting these anomalies contributed to model performances, but they also introduced critical trade-offs. Zhivkov (2021) proved that simply removing the top anomalies before training an Artificial Neural Network improves downstream supervised sensor calibration. However, directly deleting anomalies risks disrupting time-series continuity. To resolve the data loss issue, Rollo et al. (2023) introduced the anomaly repair framework using a Vector Autoregression (VAR) model to reconstruct corrupted data. However, hybrid frameworks are often too computationally expensive for real-time deployment.

Given this, we aim to see whether an unsupervised anomaly detection pipeline can identify, classify, and repair sensor faults in the UCI Air Quality dataset without discarding valid extreme pollution events? And does anomaly repair improve downstream pollutant concentration prediction compared to anomaly removal?

Data Source

We identified existing dataset sources from UCI Machine Learning Repository. The dataset contains hourly averaged responses from an array of sensors embedded in an Air Quality Chemical Multisensor Device in Italy.

Methodology

To detect abnormal sensor readings, we will implement an unsupervised anomaly detection using One - Class Support Vector Machine (OC-SVM). Initially, the UCI Air Quality dataset will be preprocessed, which includes removing missing values and normalizing pollutant measurements. This dataset contains measurements such as carbon monoxide, nitrogen oxides and other pollutants. After the preprocessing, a One-Class SVM model will be trained using historical observations. The historical data is supposed to present normal sensor behavior. The model learns what typical air quality readings look like, as One-Class SVM learns the structure of the majority of the data, by creating a boundary in feature space. Any new observation that is outside the learnt boundary will be classified as a possible anomaly. The sensor fault anomalies will be handled under two separate conditions, producing two cleaned datasets in parallel: Removal v.s. Repair. We will use linear interpolation over the nearest valid neighboring time steps, leveraging the strong temporal autocorrelation in air quality measurements.

Lastly, the model parameters will be adjusted to visualize how sensitive the anomaly detection results are to different settings. Overall, this approach allows the system to automatically detect and correct anomalous results.

Evaluation Strategies

To evaluate the real-world impact of each treatment, we will train a Random Forest model three times. First on the raw uncleaned data, second on the removal-cleaned dataset, and third on the repair-cleaned dataset. All three models will be evaluated on the same held-out test set to ensure a fair comparison to predict NO₂ concentration from the remaining sensor and meteorological features. Lastly, we will test different One-Class SVM settings to see how they affect the anomaly detection results.

References

- Zhivkov, Petar. "Optimization and Evaluation of Calibration for Low-Cost Air Quality Sensors: Supervised and Unsupervised Machine Learning Models." *Annals of Computer Science and Information Systems*, vol. 25, 26 Sept. 2021, pp. 255–258, [ieeexplore.ieee.org/document/9555721](https://doi.org/10.15439/2021f95), <https://doi.org/10.15439/2021f95>.
- Rollo, Federica, et al. "Anomaly Detection and Repairing for Improving Air Quality Monitoring." *Sensors*, vol. 23, no. 2, 6 Jan. 2023, p. 640, <https://doi.org/10.3390/s23020640>.